

Assignment 02

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Remark. We frequently use the result that if $A \subseteq B$ and $B \subseteq A$, then $A = B$. This was proven in Proposition 1 of the class 5 notes, and is in Proposition 3.1.17 of Tao. The result will be continuously used without reference.

Problem 1 (Exercise 3.1.5). *Let A, B be sets. Show that the following are equivalent.*

- (1) $A \subseteq B$.
- (2) $A \cup B = B$.
- (3) $A \cap B = A$.

Proof. (1 $\implies$ 2) Let $A \subseteq B$. Let $a \in A \cup B$. Then $a \in A$ or $a \in B$. If $a \in A$, then since $A \subseteq B$, by definition we have $a \in B$. Thus, we have $a \in B$ or $a \in B$, and thus $a \in B$. Therefore, $A \cup B \subseteq B$. Now let $b \in B$. Then $b \in B$ or $b \in A$ is true, and thus $b \in A \cup B$. Therefore, $B \subseteq A \cup B$, and thus $B = A \cup B$.

(2 $\implies$ 1) Let $A \cup B = B$. If $a \in A$, then $a \in A$ or $a \in B$, and thus $a \in A \cup B$. Since $A \cup B = B$, we have $a \in B$. Thus, $A \subseteq B$.

(2 $\implies$ 3) Let $A \cup B = B$. Let $a \in A \cap B$. Then since $a \in A$ and $a \in B$, we have $a \in A$. Thus all $a \in A \cap B$ have $a \in A$, and thus $A \cap B \subseteq A$. Conversely, let $a \in A$. To show $A \subseteq A \cap B$ and to complete this part of the proof, it suffices to show $a \in B$. By (1 $\iff$ 2), this follows.

(3 $\implies$ 1) Let $A \cap B = A$. Suppose $a \in A$. Then since $A = A \cap B$, we have $a \in A \cap B$. Thus, $a \in A$ and $a \in B$. In particular, $a \in B$, and thus $A \subseteq B$. ■

Problem 2 (Exercise 3.1.10). *Let A, B be sets.*

- (1) *Show that $A \setminus B$, $A \cap B$, and $B \setminus A$ are disjoint.*
- (2) *Show that the union of the three above sets is $A \cup B$.*

Lemma 2.1. *Let A, B be sets. If $x \in A$ implies $x \notin B$, then A and B are disjoint.*

Proof. Let $x \in A$ imply $x \notin B$. By definition, A, B are disjoint if $A \cap B = \emptyset$. Suppose for contradiction that $x \in A \cap B$. then $x \in A$ and $x \in B$. However, $x \in A$ implies that $x \notin B$, so we now have $x \in B$ and $x \notin B$, a contradiction. ■

Proof of problem 2. (1) First we show $A \setminus B$ and $A \cap B$ are disjoint. Let $x \in A \setminus B$. Then by definition, $x \in A$ and $x \notin B$. Since $a \in A \cap B$ implies $a \in B$, by contrapositive $a \notin B$ implies $a \notin A \cap B$. Thus, $x \notin A \cap B$. Thus, $x \in A \setminus B$ implies $x \notin A \cap B$. By lemma 2.1, $A \setminus B$ and $A \cap B$ are disjoint.

Now we show $A \cap B$ and $B \setminus A$ are disjoint. Let $x \in B \setminus A$. Then $x \in B$ and $x \notin A$. It follows that $x \notin A \cap B$ by the same logic as the previous step. By lemma 2.1, $B \setminus A$ and $A \cap B$ are disjoint.

Now we show $A \setminus B$ and $B \setminus A$ are disjoint. Let $x \in A \setminus B$. Then $x \in A$ and $x \notin B$. In particular, $x \notin B$. Note that $b \in B \setminus A$ implies $b \in B$ and $b \notin A$, and thus $b \in B$. By contrapositive, $b \notin B$ implies $b \notin B \setminus A$, and thus $x \notin B \setminus A$. By lemma 2.1, the sets are disjoint.

(2) First, we show $(A \setminus B) \cup (A \cap B) \cup (B \setminus A) \subseteq A \cup B$. Let $x \in (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$. Thus $x \in A \setminus B$ or $x \in A \cap B$ or $x \in B \setminus A$. We proceed in cases.

- $(x \in A \setminus B)$ We have $x \in A$ and $x \notin B$ implies $x \in A$ implies $x \in A$ or $x \in B$ implies $x \in A \cup B$.
- $(x \in B \setminus A)$ We have $x \in B$ and $x \notin A$ implies $x \in B$ implies $x \in B$ or $x \in A$ implies $x \in A \cup B$.
- $(x \in A \cap B)$ We have $x \in A$ and $x \in B$ implies $x \in A$ implies $x \in A$ or $x \in B$ implies $x \in A \cup B$.

Now we show $A \cup B \subseteq (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$. Let $x \in A \cup B$. We proceed in cases.

- $(x \in A)$ If $x \in A$, then since $x \in A \cup B$, we have two cases. If $x \in B$, then we have $x \in A$ and $x \in B$, and thus $x \in A \cap B$. If $x \notin B$, then we have $x \in A$ and $x \notin B$, and thus $x \in A \setminus B$.
- $(x \notin A)$ Since $x \in A \cup B$ and $x \notin A$, we must have $x \in B$. Thus we have $x \in B$ and $x \notin A$, which implies $x \in B \setminus A$.

■

Problem 3. Suppose that A, B, A', B' are sets such that $A' \subseteq A$ and $B' \subseteq B$.

- (1) Show that $A' \cup B' \subseteq A \cup B$ and $A' \cap B' \subseteq A \cap B$.
- (2) Give a counterexample to show that the statement $A' \setminus B' \subseteq A \setminus B$ is false. Can you find a modification of this statement involving the setminus operation which is true given the stated hypotheses?

Proof. (1) We show the statements in order. Let $x \in A' \cup B'$. Then $x \in A'$ or $x \in B'$. We proceed in cases.

- If $x \in A'$, then $A' \subseteq A$ implies $x \in A$. Thus $x \in A$ or $x \in B$, and thus $x \in A \cup B$.
- If $x \in B'$, then $B' \subseteq B$ implies $x \in B$. Thus $x \in A$ or $x \in B$, and thus $x \in A \cup B$.

Let $x \in A' \cap B'$. Then $x \in A'$ and $x \in B'$. Thus $x \in A'$. Since $A' \subseteq A$, we have $x \in A$. Additionally, since $x \in B'$ and $B' \subseteq B$, we have $x \in B$. Thus $x \in A$ and $x \in B$. Therefore $x \in A \cap B$.

- (2) Write $[n] = \{0, \dots, n\} \subseteq \mathbb{N}$. Let $A = A' = [2]$, let $B = [1]$, and let $B' = [0]$. Then

$$A \setminus B = \{2\}, \quad A' \setminus B' = \{1, 2\}.$$

There exists $1 \in A' \setminus B'$ such that $1 \notin A \setminus B$, and thus we cannot have $A' \setminus B' \subseteq A \setminus B$.

A correct statement given the hypotheses is $A' \setminus B \subseteq A \setminus B'$. Let $x \in A' \setminus B$. then $x \in A'$, and since $A' \subseteq A$, we have $x \in A$. Note that since $B' \subseteq B$, if $b \in B'$, then $b \in B$. The contrapositive of this is $b \notin B$ implies $b \notin B'$. Thus, since $x \notin B$, we have $x \notin B'$. Thus, $x \in A$ and $x \notin B'$. Therefore, $x \in A \setminus B'$. ■

Problem 4 (Exercise 3.3.5). Let

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

be functions.

- (1) If f, g are injective, show $g \circ f$ is injective.
- (2) If f, g are surjective, show $g \circ f$ is surjective.

Proof. i did this in class why do i have to do it again

(1) Let f, g be injective. Let $x_1, x_2 \in X$ such that $(g \circ f)(x_1) = (g \circ f)(x_2)$. By definition of composition, $g(f(x_1)) = g(f(x_2))$. Since g is injective, we have $f(x_1) = f(x_2)$. Since f is injective, we have $x_1 = x_2$. Therefore, $g \circ f$ is injective.

(2) Let f, g be surjective. Let $z \in Z$. Then since g is surjective, there is $y \in Y$ such that $g(y) = z$. Since f is surjective, there is $x \in X$ such that $f(x) = y$. We thus have $(g \circ f)(x) = g(f(x)) = g(y) = z$. Thus, $g \circ f$ is surjective. ■